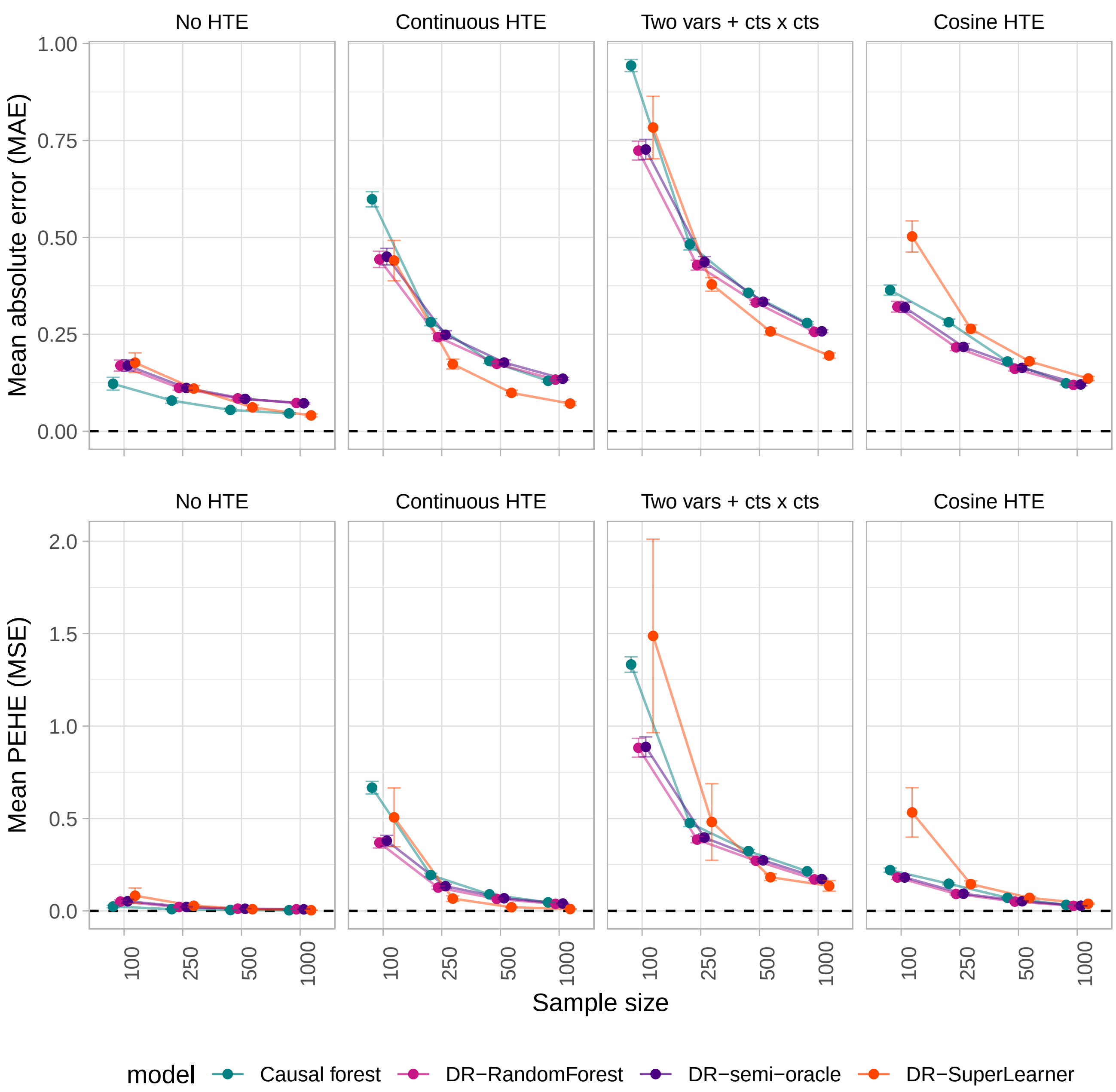
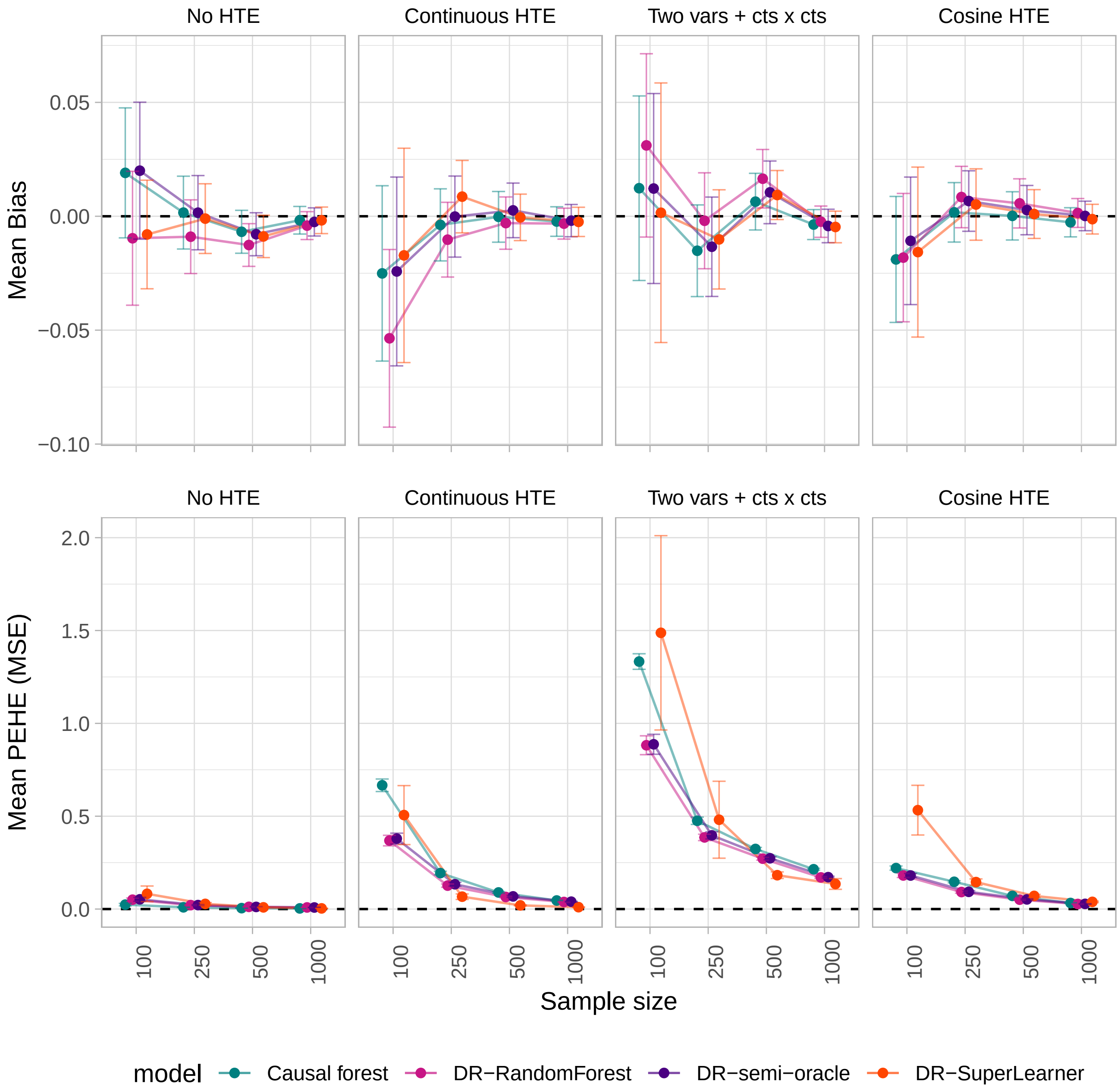
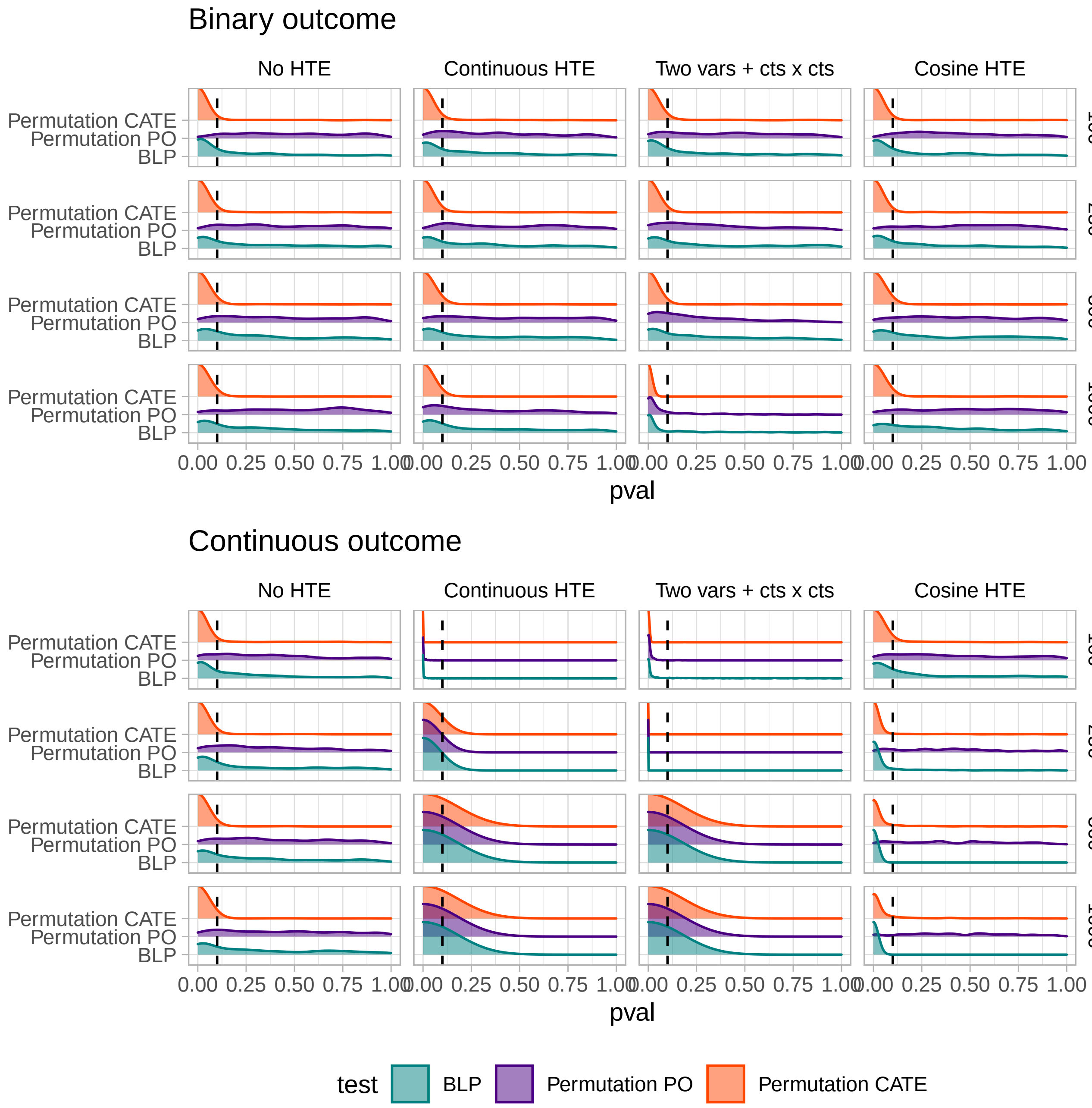
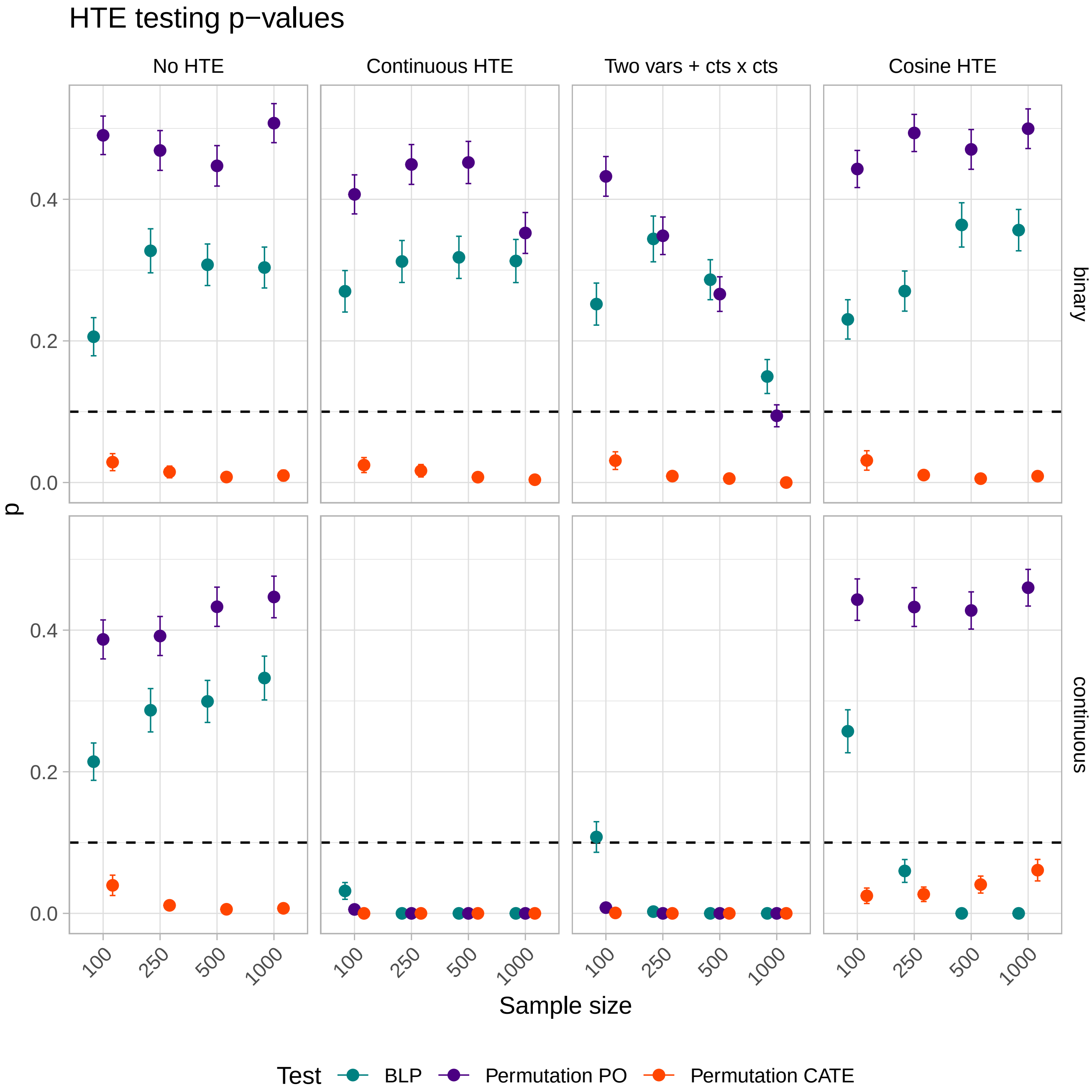

No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
Mean absolute error (MAE)
1.00
0.75
0.50
0.25
0.00
Mean PEHE (MSE)
2.0
1.5
1.0
0.5
0.0
100
250
500
1000
Sample size
model
Causal forest
DR−RandomForest
DR−semi−oracle
DR−SuperLearner

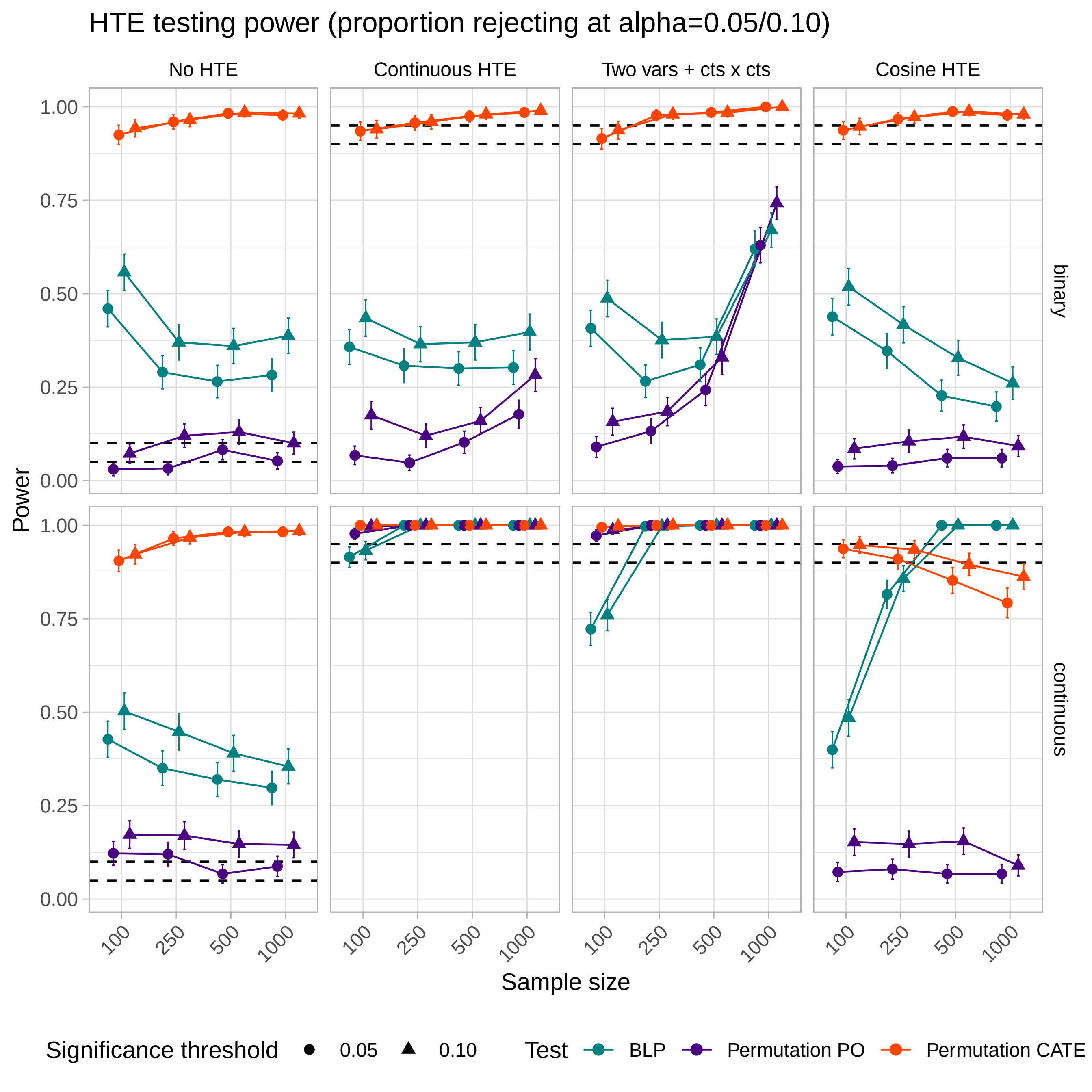


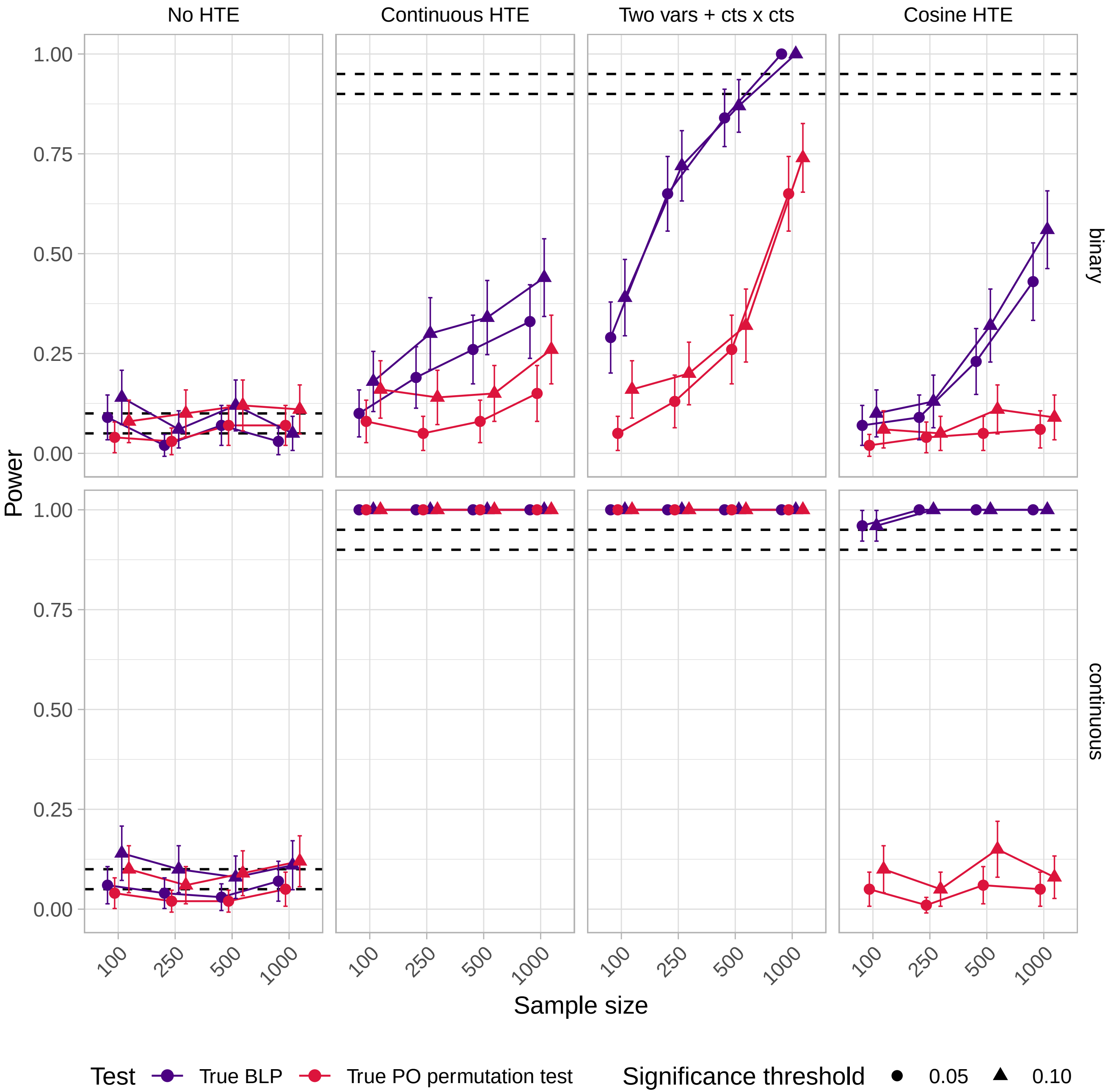
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
Mean Bias
0.05
0.00
−0.05
−0.10
Mean PEHE (MSE)
2.0
1.5
1.0
0.5
0.0
100
250
500
1000
Sample size
model
Causal forest
DR−RandomForest
DR−semi−oracle
DR−SuperLearner



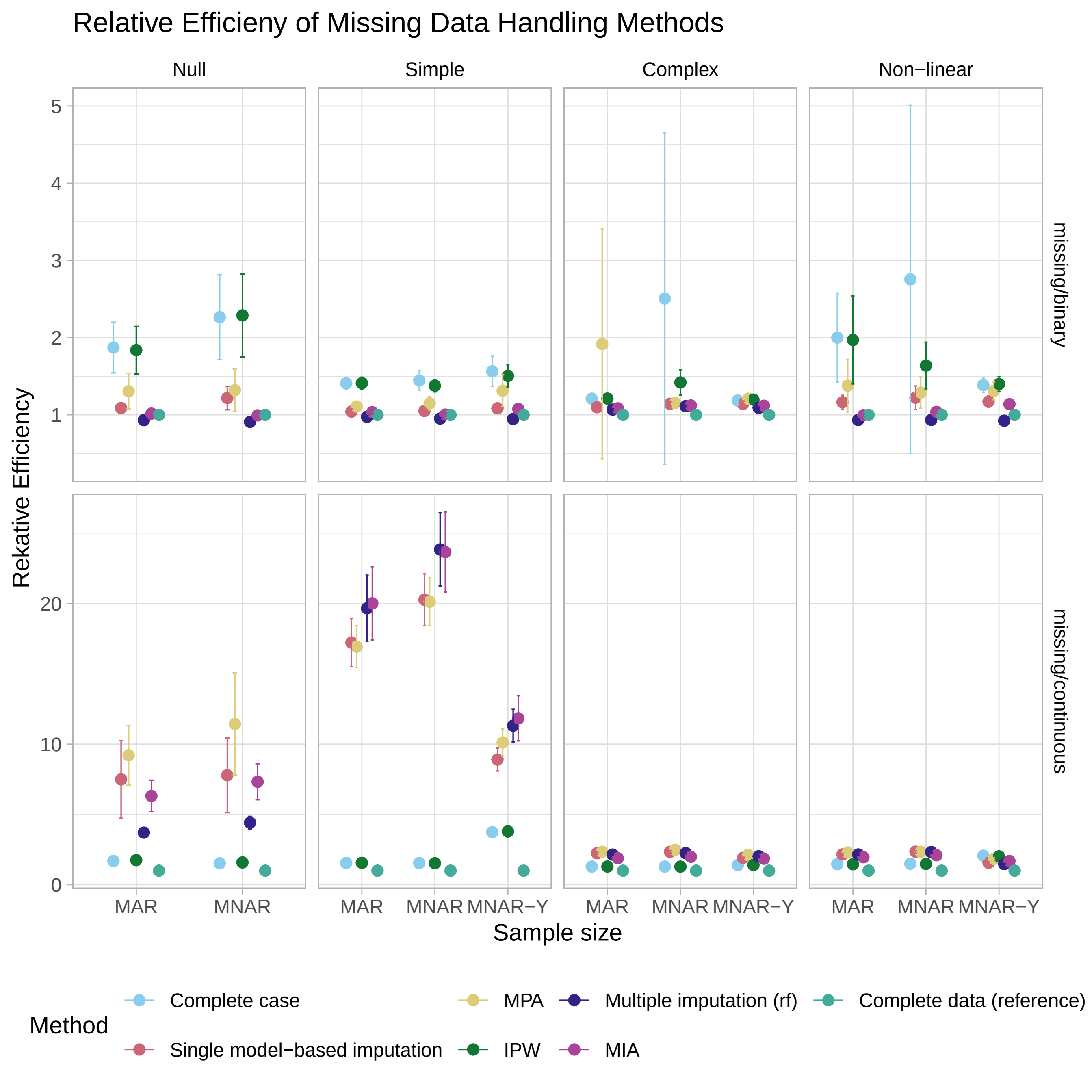
Binary outcome
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
Permutation CATE
Permutation PO
BLP
100
250
500
1000
0.00
0.25
0.50
0.75
1.00
pval
Continuous outcome
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
Permutation CATE
Permutation PO
BLP
100
250
500
1000
0.00
0.25
0.50
0.75
1.00
pval
test
BLP
Permutation PO
Permutation CATE


HTE testing p−values
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
binary
continuous
p
0.0
0.2
0.4
100
250
500
1000
Sample size
Test
BLP
Permutation PO
Permutation CATE

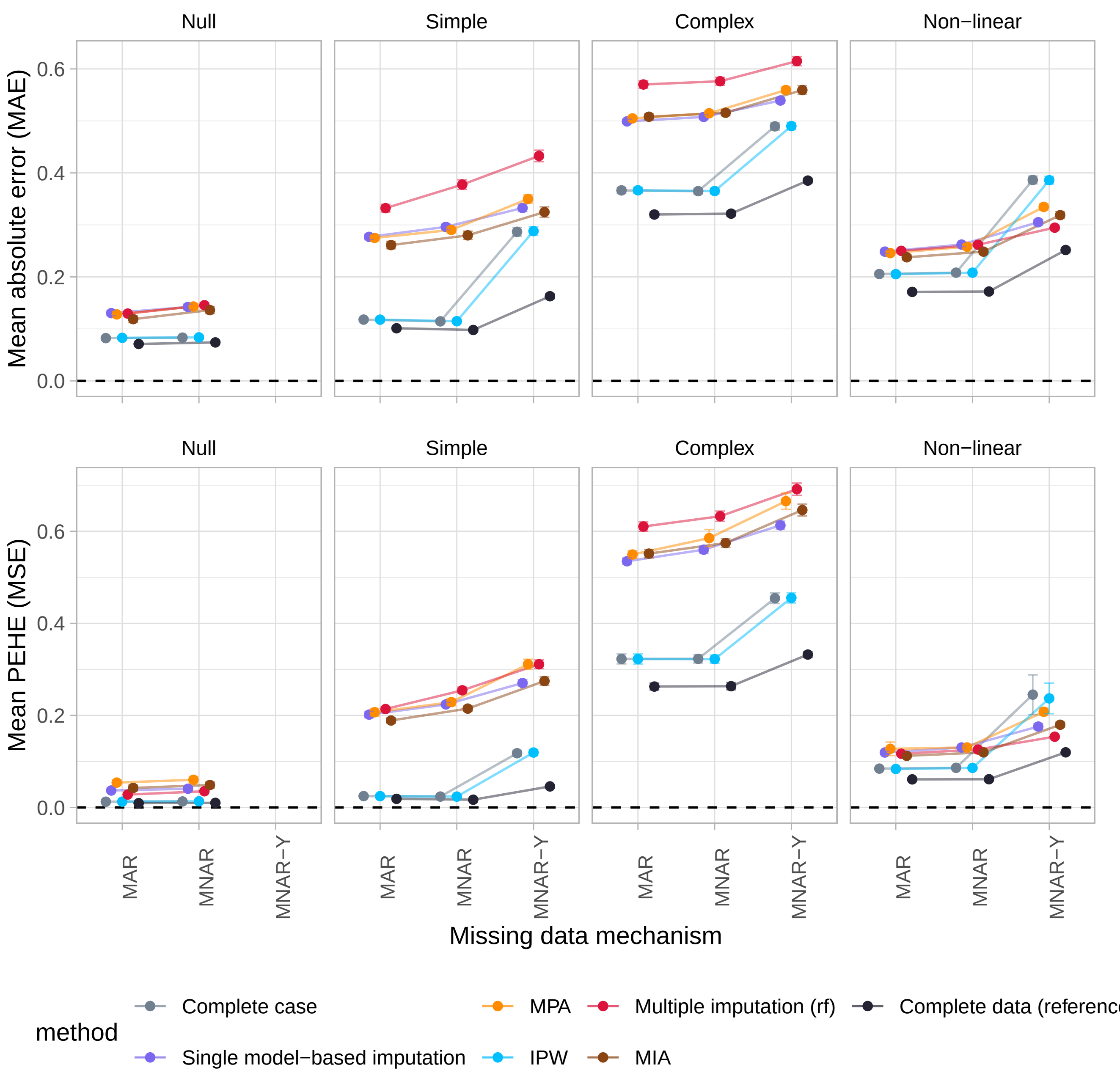

HTE testing power (proportion rejecting at alpha=0.05/0.10)
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
binary
continuous
Power
0.00
0.25
0.50
0.75
1.00
100
250
500
1000
Sample size
Significance threshold
0.05
0.10
Test
BLP
Permutation PO
Permutation CATE



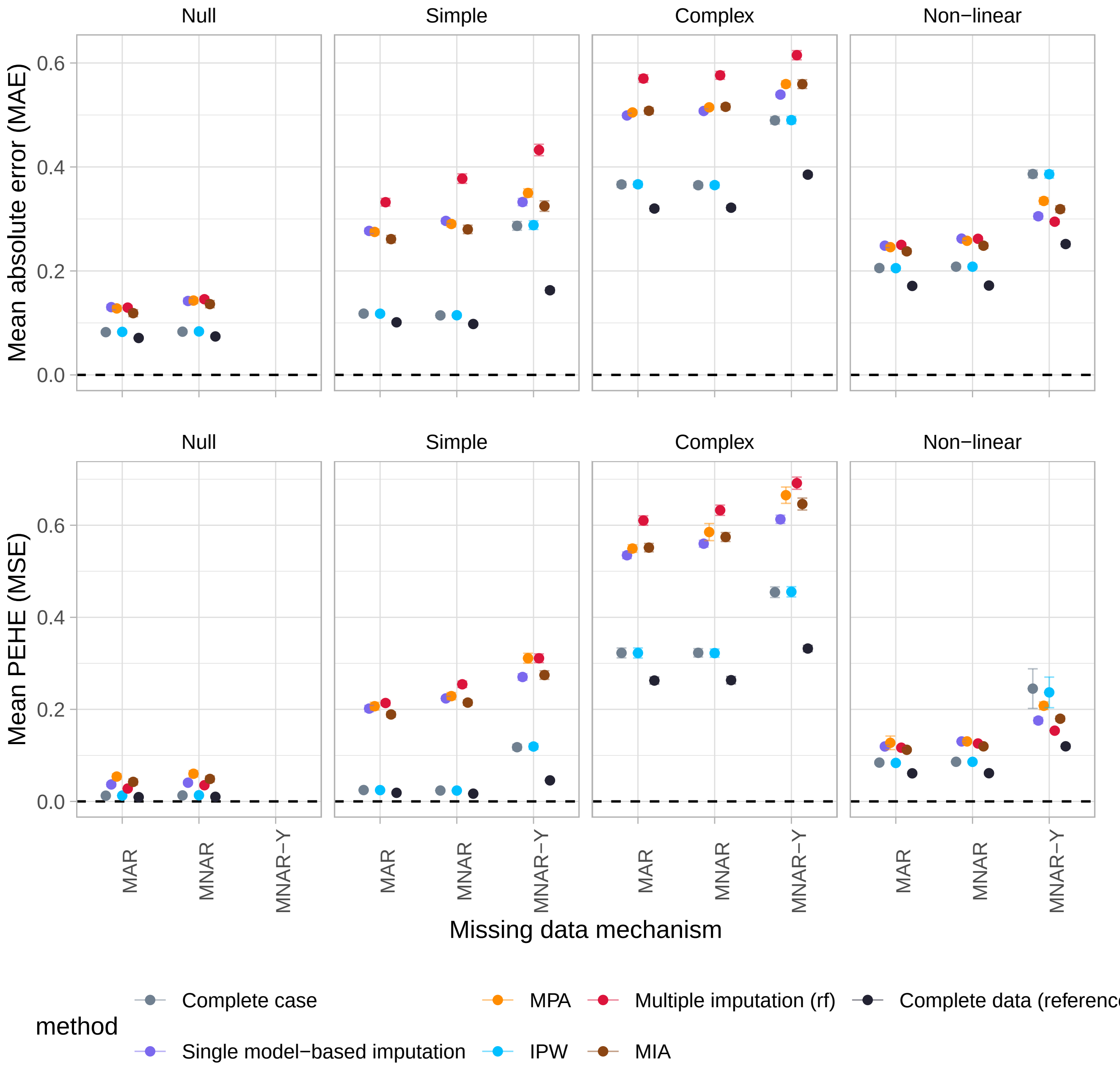
No HTE
Continuous HTE
Two vars + cts x cts
Cosine HTE
binary
continuous
Power
1.00
0.75
0.50
0.25
0.00
100
250
500
1000
Sample size
Test
True BLP
True PO permutation test
Significance threshold
0.05
0.10



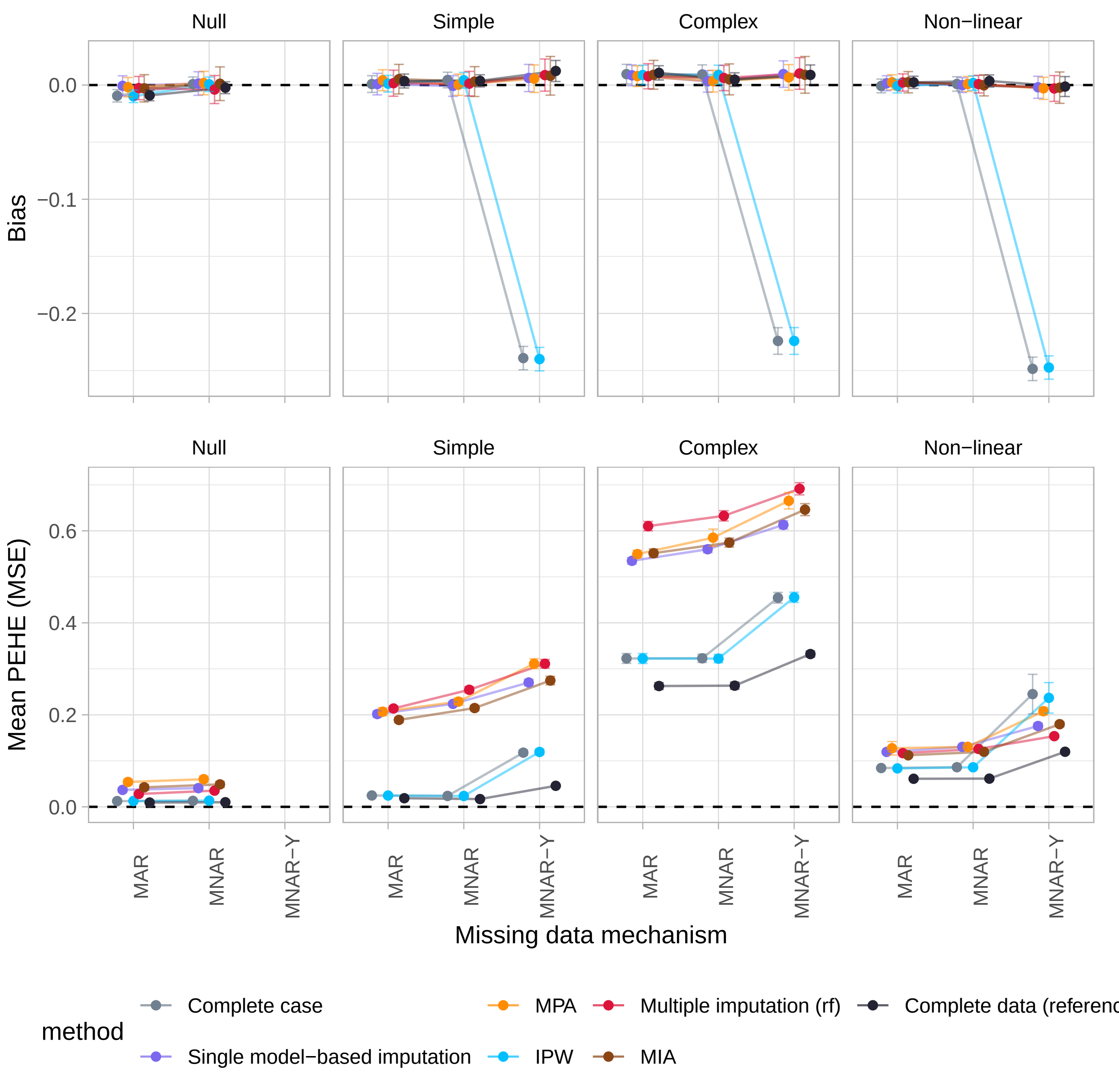
Relative Efficieny of Missing Data Handling Methods
Null
Simple
Complex
Non−linear
missing/binary
missing/continuous
Rekative Efficiency
Sample size
MAR
MNAR
MNAR−Y
Method
Complete case
Single model−based imputation
MPA
IPW
Multiple imputation (rf)
MIA
Complete data (reference)



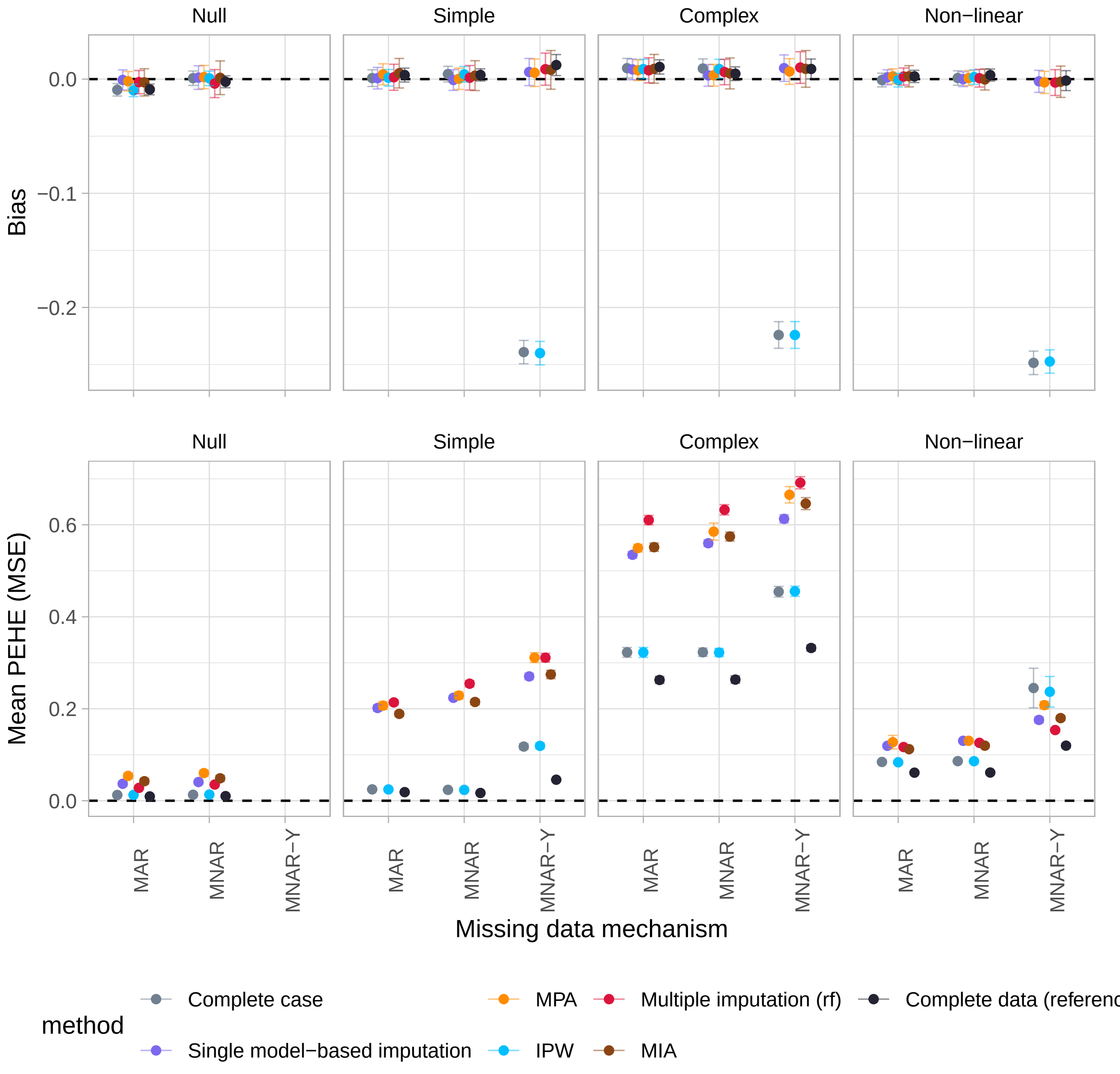
Null
Simple
Complex
Non−linear
Mean absolute error (MAE)
Mean PEHE (MSE)
0.0
0.2
0.4
0.6
MAR
MNAR
MNAR−Y
Missing data mechanism
method
Complete case
Single model−based imputation
MPA
IPW
Multiple imputation (rf)
MIA
Complete data (referenc



Null
Simple
Complex
Non−linear
Mean absolute error (MAE)
0.0
0.2
0.4
0.6
Mean PEHE (MSE)
MAR
MNAR
MNAR−Y
Missing data mechanism
method
Complete case
Single model−based imputation
MPA
IPW
Multiple imputation (rf)
MIA
Complete data (referenc



Null
Simple
Complex
Non−linear
Bias
0.0
−0.1
−0.2
Mean PEHE (MSE)
0.6
0.4
0.2
0.0
MAR
MNAR
MNAR−Y
Missing data mechanism
method
Complete case
Single model−based imputation
MPA
IPW
Multiple imputation (rf)
MIA
Complete data (referenc



Null
Simple
Complex
Non−linear
Bias
0.0
−0.1
−0.2
Mean PEHE (MSE)
0.6
0.4
0.2
0.0
MAR
MNAR
MNAR−Y
Missing data mechanism
method
Complete case
MPA
Multiple imputation (rf)
Complete data (referenc
Single model−based imputation
IPW
MIA